Regulatory Capture via Price Aggregator

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Motivation

Large firms support regulations that seem costly for them:

- Meta: Zuckerberg asked Congress for tighter content-moderation rules.
- Amazon: pushing for minimum wage increases.
- Microsoft: backed 'clean grid' computing legislation.
- ExxonMobil: lobbyists pushing for sub-2°C policies.

Standard intuition:

- Political pressure / signaling.
- Getting exemptions / enforcement advantages.

This paper:

- Even when a regulation increases everyone's prices, if burden depends on productivity, relative prices decrease for the most productive.
- This leads to higher market shares and profits.

This paper:

- Regulations may look neutral, but have an asymmetric impact across firms. E.g., minimum wage and size-dependent factor shares.
- Large/productive firms benefit from costly regulation, as long as it reduces their relative price.
- A reduction in relative price leads them to higher market share and profit.

Intuition: firms are willing to pay for regulation that hurts them, as long as it hurts their competitors more.

The regulation class we study

Target: industry-wide rules that raise marginal costs, but whose marginal burden collapses for large/productive firms.

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Amazon example:

- 2012: Acquires Kiva Systems (now Amazon Robotics, automation arm).
- 2016: Starts integrating robotics in fulfillment centers (min wage workers).
- **2018: Push for doubling the federal min wage.**
- 2022: Massive layoffs start (mainly in fulfillment centers).
- 2025: Internal leaked documents suggest that automation is going to save them 600,000 jobs over the next decade.

Outline

- Place in the literature.
- Theoretical framework.
  - Endogenous market share and profit reallocation from exogenous asym. burden.
  - Endogenous asym. burden through markups, and/or
  - Regulator side.
- Special Case: CES with Pareto costs.
- Estimation.
- Counterfactuals.

Literature Review

This paper sits at the intersection of three literatures:

- **Regulation as a traded good/regulator as agent (micro/GT).**
  Seminal: **?**, **?**, **?**; Survey: **?**; Recent synthesis/empirics: **?**.
- **Raising Rivals' Costs, RRC (IO/PE).**
  Canonical: **??**, **?**, **?**. Heterogeneous compliance in environmental settings: **?**.
- **Firm size and capture (applied/macro).**
  Theory emphasizing fixed-cost channels: **?**.
  Empirics on connected/large firms: **?**, **?**, **?**.

Preview: the contribution is showing how **uniform, industry-wide** rules can generate rents within an industry **without needing exemptions, discriminatory enforcement, or inducing exit**.

→ Detailed literature review: key assumptions and what we take from each.

Raising Rivals' Costs vs Capture via Aggregator

	Canonical RRC	**This Paper**
Market structure	Oligopoly with **homogeneous goods**	Monopolistic competition
Strategic interaction	Essential (**Cournot/Stackelberg**)	Not required
Type of regulation	Targeted or asymmetric incidence	Formally neutral, industry-wide
Source of advantage	Exemptions, enforcement differences, entry barriers	**Heterogeneous effective burden**

Contributions

1. **A new capture channel via price aggregator.** Industry-wide marginal cost increases can be privately beneficial for the most productive firms if they shift the price aggregator enough that their **relative price falls**.
2. **Beyond CES.** The logic holds for a broad class of homothetic demand systems; CES appears only as a special case.

Roadmap: next we formalize why the relative price is the correct object of interest, then define regulation and classify them by how they deform the marginal cost distribution.

Theoretical Framework

Environment

- Continuum of differentiated goods $\omega \in (0, 1)$, in static closed economy.
- Consumers represented by a homothetic demand system with price aggregator $A(\mathbf{p})$:

$$x_\omega = x_\omega \left(\frac{p_\omega}{A(\mathbf{p})} \right)$$

- Heterogeneous firms in monopolistic competition with marginal cost $\psi_\omega > 0$ maximize:

$$\pi_\omega = (p_\omega - \psi_\omega) x_\omega$$

Environment

Using HSA demand systems (**?**), you can rewrite the firms maximization problem the following way:

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Where $s_\omega(\cdot)$ is the share function, Y is expenditure, and $A(\cdot)$ a homothetic aggregator. Finally $z_\omega \equiv \frac{p_\omega}{A(\mathbf{p})}$ is the relative price.

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Key Implication: Maximizing over price levels p_ω is equivalent to maximizing over relative prices z_ω, and equilibrium profit and shares are decreasing in the **relative marginal cost** $\boldsymbol{\frac{\psi_\omega}{A(\mathbf{p})}}$: $\frac{\partial \pi_\omega^*}{\partial \psi_\omega / A(\mathbf{p})} < 0$.

When can higher costs help a firm?

From $\frac{\partial \pi_\omega^*}{\partial \psi_\omega / A(\mathbf{p})} < 0$ and $\mathbf{p}^* = \boldsymbol{\mu\psi}$, firm ω benefits from a regulatory change in marginal costs whenever its **relative marginal cost falls**:

$$\frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

even if its own marginal cost rises, $\psi_\omega^R > \psi_\omega$.

Regulations

Suppose marginal costs come from a family of continuous parametric CDFs, $\psi \sim G \in \mathcal{G}$.

Def. Regulation:

A regulation changes marginal costs to $\psi^R \sim G^R$, with $G, G^R \in \mathcal{G}$.

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Why parameterize? It lets us classify regulations by the way they deform the distribution: **scale**, **location**, or **shape**.

Changes in the shape parameter will lead to higher marginal costs for everyone, but will affect firms asym. lees the more productive they are.

E.g., in Pareto costs: $G(\psi) = 1 - \left(\frac{\psi_0}{\psi}\right)^{\kappa}$, κ is the shape parameter.

Shape Deformations

Def. R-cis:

A policy R is cost increasing in shape if for every firm $\omega \in (0,1)$ the marginal cost increases through a change in the shape parameter. That is, if $\psi_\omega^R > \psi_\omega \ \forall \ \omega \in (0,1)$ (a stricter version of FOSD), through a change in $\boldsymbol{\kappa}$.

Thm 1:

The most productive firms always benefit from R-cis.

→ Minimal statement for non parametric ψ, no dominance between ψ and ψ^R, and/or no shared family $\mathcal{G}$.

→ Adding choke prices increases benefits for the most productive firms (that survive).

Proof of Thm 1

Proof of Thm 1: The most productive firms always benefit from R-cis.

- Since $\psi_\omega^R > \psi_\omega \ \forall \omega \in (0,1)$, $A(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi})$.
- By continuity of $\mathcal{G}$, and since κ is support invariant:

$$\lim_{\omega \downarrow 0} \frac{\psi_\omega^R}{\psi_\omega} = 1$$

This implies that there exists $\varepsilon > 0$ so that $\forall \omega \in (0, \varepsilon)$:

$$\frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

■

Example: CES demand with $\psi \sim$ Pareto.

Setting

Recall that firms pick relative price z_ω as to maximize:

$$\pi_\omega = \left(1 - \frac{\psi_\omega / A(\mathbf{p})}{z_\omega}\right) Y s_\omega \left(z_\omega\right)$$

Since we have a CES demand system,

$$A(\mathbf{p}^*) = A(\mu \boldsymbol{\psi}) = \left[\int_0^1 \left(\frac{\sigma}{\sigma - 1}\psi_\gamma\right)^{1-\sigma} \partial\gamma\right]^{\frac{1}{1-\sigma}}.$$

Relative Marginal Cost

Assume that $\psi \sim \text{Pa}(\psi_0, \kappa)$, where ψ_0 and κ are the scale and shape parameters, respectively.

The relative marginal cost for firm ω would then be:

$$\frac{\psi_\omega}{\mu A(\boldsymbol{\psi})} = \frac{(1-\omega)^{-\frac{1}{\kappa}}}{\mu \left[\int_0^1 (1-\gamma)^{\frac{\sigma-1}{\kappa}} \partial\gamma \right]^{\frac{1}{1-\sigma}}}$$

$$= \frac{(1-\omega)^{-\frac{1}{\boldsymbol{\kappa}}}}{\mu \left[\frac{\boldsymbol{\kappa}}{\boldsymbol{\kappa}+\sigma-1} \right]^{\frac{1}{1-\sigma}}}$$

R that marginally decreases κ

Suppose there is a regulation R that **marginally** decreases the shape parameter κ. This would imply that $\psi_\omega^R > \psi_\omega \ \forall \omega \in (0,1)$. According to **Thm 1**, we can find $\varepsilon > 0$ so that every firm in $(0, \varepsilon)$ is benefited. In particular a firm ω benefits if:

$$\frac{\partial \ln(\psi_\omega / A(\boldsymbol{\psi}))}{\partial \kappa} > 0$$

This implies that the only firms that would benefit are $\omega \in (0, \varepsilon)$ with:

$$\varepsilon = 1 - \exp\left(-\frac{\kappa}{\kappa + \sigma - 1}\right)$$

From **Thm 1**, any decrease in any initial κ would benefit the most productive firms.

Other Deformations

Scale Deformations (benchmark neutrality)

Scale changes correspond to proportional compliance wedges, regulations that are isomorphic to ad valorem taxes or uniform input price increases.

Def. R-cic:

A policy R is cost increasing in scale if for every firm $\omega \in (0,1)$ the marginal cost increases through a change in the scale parameter. That is, $\psi_\omega^R = r\psi_\omega \ \forall \ \omega \in (0,1)$ and $r > 1$.

Thm 2:

Firms are neither affected or benefited by R-cic.

This pins down a clean boundary: if regulation is proportional, there is no rent generated from $A(\cdot)$.

Thm 2

Thm 2: Firms are neither affected or benefited by R-cic.
Changes in the scale parameter multiply every marginal cost by the same constant r, so the only question is whether the market index scales in the same way. In the HSA setting it does, by homogeneity of $A(\cdot)$.:

$$\frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)} = \frac{r\psi_\omega}{A(r\boldsymbol{\mu\psi})} = \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

Location Shifts (the opposite boundary)

The remaining deformation is a parallel shift: every firm pays the same absolute increment in marginal cost. Regulations that are isomorphic to fixed per-unit fees.

Def. R-cil:

A policy R is cost increasing in location if for every firm $\omega \in (0,1)$ the marginal cost increases through a change in the location parameter. That is, $\psi_\omega^R = \psi_\omega + a \ \forall$ $\omega \in (0,1)$ and $a > 0$.

Thm 3:

The most productive firms are negatively affected by R-cil.

So: increases in scale are neutral, in location are anti-top-firm, and in shape is where pro-top-firm regulation arises.

Thm 3

Thm 3: The most productive firms are negatively affected by R-cil.
A location shift adds the same absolute increment a to every firm. That implies percentage cost increases are **larger** for low-cost firms. HSA regularity, continuity of $\mathcal{G}$, and IVT imply there exists a cutoff $\varepsilon \in (0,1)$ that is exactly indifferent:

$$\frac{A(\boldsymbol{\mu}(\boldsymbol{\psi}+a))}{A(\boldsymbol{\mu}\boldsymbol{\psi})} = \frac{\psi_\varepsilon + a}{\psi_\varepsilon}$$

Then for every $\omega \in (0,\varepsilon)$ the proportional increase is larger than at ε, so relative marginal cost rises:

$$\frac{\psi_\varepsilon + a}{\psi_\varepsilon} < \frac{\psi_\omega + a}{\psi_\omega} \iff \frac{\psi_\omega}{A(\boldsymbol{\mu}\boldsymbol{\psi})} < \frac{\psi_\omega + a}{A(\boldsymbol{\mu}(\boldsymbol{\psi}+a))}$$

This implies the most productive firms are always negatively affected by R-cil.

→ Proof of Thm 3 (existence of ε).

Conclusion

- **Mechanism:** for a broad class of homothetic demands, firm outcomes are determined by **relative price**. Regulation can shift $A(\cdot)$ enough to increase market share and profits for the most productive firms.
- **Main result (Thm 1):** asym. increases in marginal cost benefit the most productive firms. Equilibrium prices increase, but relative prices decrease which leads to higher market share and profits.
- **Special Case:** in CES with Pareto costs, a marginal decrease in κ delivers closed form expression for the cutoff ε.
- **Boundary conditions:** proportional increases are **neutral** by homogeneity (Thm 2); additive shifts **hurt** top firms (Thm 3).
- **Implication:** capture need not rely on exemptions, discriminatory enforcement or induce exit—**regulation can still redistribute** through the aggregator.

Appendix Index

- Detailed Literature Review.
- Thm 1: Minimal.
- Choke Prices.
- Proof of Thm 3.
- References.

Literature Review

1 Regulation as a traded good / regulator as agent (micro/GT).

- **Core idea:** regulation is endogenous; organized interests demand it; policymakers supply it.
- **Seminal:** **?**, **?**, **?**, **?**, **?**.
- **Common-agency benchmark:** **?** (contribution schedules; Nash equilibrium).
- **Heterogeneous lobbying inside industries:** **?**.
- **Survey / synthesis:** **?**; **??**.
- **What we add:** a new **within-industry conflict object** under a uniform instrument:

$$\text{winners/losers are pinned down by } \frac{\psi_\omega}{A(\mu\psi)} \text{ rather than by } \psi_\omega.$$

In many canonical political-economy settings, incentives within an industry are aligned and the competition is between different industries.

2 Raising Rivals' Costs, RRC (IO/PE).

Early evidence: ? (economic history; no formal model): first enforced social policy (child labor laws) as capture, to strategically increase costs.

RRC canonical modeling choices:

- **Formal RRC: ??**: regulations, standards, advertising rules, input contracts that raise rivals' costs. Finite-firm oligopoly; partial equilibrium; strategic interaction.
- Entry deterrence / exclusion: **?**.
- Raising rivals' fixed costs / inducing exit: **?**.

How our mechanism differs:

- RRC advantage requires asymmetric incidence: exemptions, enforcement differences, or cost shocks that hit rivals more.
- Here: even **uniform, industry-wide marginal-cost** regulation can create winners via shifts in the aggregator $A(\cdot)$.

Related "policy environments" where incidence is heterogeneous: **?**. Related empirical IO on own vs rivals' cost pass-through: **?**.

3 Firm size and capture (applied/macro).

Modern theory: large firms favor stricter regulation

- **?** (trade setting; CES, Pareto productivity): large firms can favor regulation that **increases fixed costs** because it induces exit of small rivals and raises markups.
- Knife-edge case according to this setting: their logic implies **neutrality of marginal-cost regulation** (scale changes), consistent with Thm 2.
- **Our distinction:** variable-cost regulation can matter even without targeted incidence if it changes the **shape** of the marginal-cost distribution (Thm 1).

Consistent Evidence:: size/connected firms gain in regulated sectors

- **?**: valuation/markup/entry patterns for connected firms in regulated sectors.
- **?**: markups and political connections.
- **?**: lobbying and firm size in a macro GE setting with idiosyncratic shocks to lobbying firms.

←

Thm 1: Minimal

Thm 1: Minimal.

If the following two conditions hold:

1. Strict index ordering:
$$A(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi})$$
2. Left-tail convergence to 1:
$$\lim_{\omega \downarrow 0} \frac{\psi_\omega^R}{\psi_\omega} = 1$$

Then there exists $\varepsilon > 0$ so that $\forall \omega \in (0, \varepsilon)$:

$$\frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

←

Choke Prices

Choke Prices

Choke prices are not required for capture via $A(\cdot)$; they act as an amplifier by inducing exit.

As in **?**, let the choke prices be $\bar{p}_\omega = \bar{z}_\omega A(\mathbf{p}) > 0$; $s_\omega(\cdot)$ the primitive for HSA demand systems; and the HSA price aggregator $\boldsymbol{A(\cdot)}$ **implicitly defined** by:

$$\int_\Omega s_\omega \left(\frac{\mu_\omega \psi_\omega}{A(\boldsymbol{\mu\psi})} \right) \partial\omega = 1 \text{ for all possible } \boldsymbol{\mu\psi}$$

Without choke prices, we do not need to think about changes in the functional form of $A(\cdot)$, since it is already defined for all possible $\boldsymbol{\mu\psi}$. But since choke prices are going to induce exit, this will change the measure of firms Ω that participate in the market (thus implicitly changing the definition of $A(\cdot)$).

Choke Prices

Suppose a regulation R-cis induces a positive measure of firms to exit, $|\Omega| > |\Omega^R|$ (i.e., choke prices are sufficiently binding). $A^R(\cdot)$ is then implicitly defined as:

$$\int_{\boldsymbol{\Omega^R}} s_\omega \left(\frac{\mu_\omega \psi_\omega}{A^R(\boldsymbol{\mu\psi})} \right) \partial\omega = 1 \text{ for all possible } \boldsymbol{\mu\psi}$$

Since the (decreasing in their own argument) consumers $s_\omega(\cdot)$ do not change, $|\Omega| > |\Omega^R|$, and

$$\int_{\Omega} s_\omega \left(\frac{\mu_\omega \psi_\omega}{A(\boldsymbol{\mu\psi})} \right) \partial\omega = \int_{\boldsymbol{\Omega^R}} s_\omega \left(\frac{\mu_\omega \psi_\omega}{A^R(\boldsymbol{\mu\psi})} \right) \partial\omega \text{ for all possible } \boldsymbol{\mu\psi}$$

Then it must be true that $A^R(\boldsymbol{\mu\psi}) > A(\boldsymbol{\mu\psi})$ for all possible $\boldsymbol{\mu\psi}$.

Intuition: with fewer active firms and unchanged demand primitives, the price index must rise to re-normalize shares.

Choke Prices

Cor 1: Exit as an amplifier.

When choke prices bind, the most productive surviving firms reap strictly larger benefits from R-cis.

Proof.
From **Thm 1**, in the absence of binding choke prices, $A(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi})$ and $\exists\ \varepsilon > 0$ so that every firm in $(0, \varepsilon)$ benefits.
From above, if choke prices are binding, then $A^R(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi}^R)$, so that for all $\omega \in (0, \varepsilon) \cap \Omega^R$:

$$\frac{\psi_\omega^R}{A^R(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)}$$

■

Unlike standard capture stories, exit is not necessary, but instead an amplifier of a mechanism that already operates through the price aggregator $A(\cdot)$.

←

Proof. Thm 3

Proof of Thm 3 (existence of ε)

Thm 3: The most productive firms are negatively affected by R-cil.

Let $\psi_\omega = G^{-1}(\omega)$ be continuous, strictly increasing, with $\psi_\omega \geq \psi_0 > 0$ and $\psi_\omega \to \infty$ as $\omega \uparrow 1$. Let $A(\cdot)$ be monotone and homogeneous of degree 1.
Define

$$\rho(a) \equiv \frac{A(\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1}))}{A(\boldsymbol{\mu}\boldsymbol{\psi})}, \qquad a > 0.$$

Step 1 (bounds on $\rho(a)$). Monotonicity implies $\rho(a) > 1$ since $\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1}) > \boldsymbol{\mu}\boldsymbol{\psi}$ componentwise. Also, since $\psi_\omega \geq \psi_0$,

$$\psi_\omega + a \leq \left(1 + \frac{a}{\psi_0}\right)\psi_\omega \;\; \forall \omega \Rightarrow A(\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1})) \leq \left(1 + \frac{a}{\psi_0}\right)A(\boldsymbol{\mu}\boldsymbol{\psi}),$$

where the last step uses monotonicity and homogeneity. Hence $\rho(a) \in \left(1,\, 1 + \frac{a}{\psi_0}\right]$.

Proof of Thm 3 (existence of ε)

Step 2 (IVT). Let $g(\omega) = \frac{\psi_\omega + a}{\psi_\omega} = 1 + \frac{a}{\psi_\omega}$. Then g is continuous and strictly decreasing, with

$$\lim_{\omega \downarrow 0} g(\omega) = 1 + \frac{a}{\psi_0}, \qquad \lim_{\omega \uparrow 1} g(\omega) = 1.$$

Since $\rho(a) \in (1,\, 1 + \frac{a}{\psi_0}]$, by IVT there exists a unique $\varepsilon \in (0, 1)$ such that $g(\varepsilon) = \rho(a)$, i.e.

$$\frac{A(\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1}))}{A(\boldsymbol{\mu}\boldsymbol{\psi})} = \frac{\psi_\varepsilon + a}{\psi_\varepsilon}.$$

For $\omega < \varepsilon$, $g(\omega) > g(\varepsilon)$, hence $\frac{\psi_\omega + a}{A(\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1}))} > \frac{\psi_\omega}{A(\boldsymbol{\mu}\boldsymbol{\psi})}$, so top firms are harmed. ■

←

The regulation class we study (and that the literature largely ignores)

Target: industry-wide rules that raise marginal costs, but whose marginal burden collapses for large/productive firms.

- **Output-linked reporting / continuous evidence regimes**
  Examples: MiFID II transaction reporting; SEC CAT; VAT real-time invoicing (Brazil SPED; Italy SDI) / NERC CIP continuous compliance.
- **Tail-event constraints**
  DOT tarmac delay, EPA RMP accident prevention, PHMSA pipeline integrity management.
- **Endogenous financial assurance**
  UST financial responsibility, RCRA hazardous waste financial assurance, oil spill COFR, mining reclamation bonds (SCMRA), freight broker bond (FMCSA).
- **Discrete compliance thresholds**
  Two-stair requirement.

These regulations are uniform in design, but heterogeneous in **effective incidence** across the productivity distribution.

References I